MILITARY SPECIFICATION

NAVIGATIONAL SET, OMEGA
AN/ARN-99(V)1

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive the three-frequency OMEGA transmissions and transmit, in digital form, the sine and cosine correlations of these signals to an associated Computer.

1.2 Classification - The equipment covered in this specification shall consist of the following items:

Units	Type Designation	Applicable Paragraph
(1) Receiver-Converter	OR-90/ARN-99(V)	3.5.1
(2) Antenna-Coupler	AS-2623/ARN-99(V) or AS-2623A/ARN-99(V)	3.5.2
(3) Mounting Base	MT-4534/ARN-99(V)	3.5.3

1.3 Associated Equipment - The equipment covered in this specification shall operate with the typical associated equipment listed in Paragraph 6.7.
The performance of the OMEGA Navigational Set AN/ARN-99(V)1 with typical associated equipment shall be as specified in Appendix A.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088 Wiring; Aircraft, Installation of

MIL-E-5400 Electronic Equipment, Aircraft, General Specification for

FSC 5826

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

MIL-T-5422 Testing, Environmental, Aircraft Electronic Equipment

MIL-M-7793 Meter, Time Totalizing

MIL-E-17555 Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

MIL-T-18303 Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for

MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-T-23103 Thermal Performance Evaluation, Airborne Electronic Equipment; General Requirement for

MIL-C-81332 Computer Set, Digital AN/ASQ-114(V)

MIL-C-81511 Connectors, Electrical, Circular, High Density Quick Disconnect, Environment Resisting; and Accessories General Specification for

Naval Air System Command

AR-5 Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of

AR-10 Maintainability of Avionics Equipment and Systems, General Requirements for

AR-34 Failure Classification for Reliability Testing, General Requirements for

WR-101 Weapons Requirement for Advanced ASW Avionics System Part II - Selection and Testing of Microelectronic Circuitry for Advanced ASW Avionics Systems; A-NEW

STANDARDS

Military

MIL-STD-415 Test Provisions for Electronic System and Associated Equipment, Design Criteria for

MIL-STD-461 Electromagnetic Interference Characteristics, Requirements for Equipment

MIL-STD-470 Military Standard Maintainability Program Requirements

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

MIL-STD-471 Military Standard Maintainability Demonstration

MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

MIL-STD-781 Reliability Tests, Exponential Distribution

MIL-STD-785 Reliability Program for System and Equipment Development and Production

MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

MS-17322 Meter, Time Totalizing, Miniature, Digital, 115 Volt, 400 Hertz

Federal

FED-STD-595 Colors

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic Technology shall be considered, and microelectronic items shall conform to requirements specified herein. Part II of WR-101 shall be used as a guide.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10. Plug in subassemblies shall be so designed as to preclude improper interchange or insertion.

3.2.4 AR-10 Deviations - WRAs (Weapon Replaceable Assembly) shall not be equipped with fault location indicators as required by Paragraph 3.3.1.2 of AR-10. The fault location indicator for the Receiver-Converter Group shall be a display of the associated display unit under associated computer control. See also Paragraph 3.3.7.9.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 37.85 pounds.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 3000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 20,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment, including any Built-in-Test provisions, shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of Paragraph 4.4.3.

3.3.2.4 Time Totalizing Meter - The Receiver-Converter Group shall include a time totalizing meter conforming with the applicable requirements of Specification MIL-M-7793 and MS-17322-10. Meter shall be located on Interconnecting Box J-3008/ARN-99(V).

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. Miniature quick disconnect circular connectors shall conform to the applicable requirements of Specification MIL-C-81511.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - The control will be provided as specified in Appendix A.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400. No adjustments shall be required when replacing plug-in subassemblies.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, except for the following modifications:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

Receiver-Converter

CE01 Broadband	Use of 2.5kHz High Pass Filter
CE01 Narrowband	30 db above specification limit
RE01	20 db above specification limit

Antenna-Coupler

RS02 - Test levels for the case test shall be limited to the threshold of antenna susceptibility.

3.3.7 Provisions for Maintainability - Built-in test equipment, construction and packaging, and provisions for test points shall be as specified herein and generally in accordance with Specification AR-10.

3.3.7.1 Compatibility with VAST - The equipment shall conform with the maintainability philosophy as specified herein. VAST compatibility is desirable, but is not a prime design objective.

3.3.7.2 Maintainability Program - The contractor shall conduct a maintainability program in accordance with MIL-STD-470.

3.3.7.3 Maintainability Requirements - The equipment shall be such that all corrective maintenance can be accomplished by maintenance personnel who have attained the E-4 skill level and who have successfully completed appropriate training programs.

3.3.7.4 Organizational Maintainability - The characteristics of the equipment shall be such that it will be possible in 90% of the cases of failure, to perform all corrective organization maintenance actions, other than combat damage repair, within a period not exceeding 30 minutes (the 30 minute corrective action time shall include all elements except aircraft access time and time to obtain spares). The 30 minute corrective action time is predicated on fault isolation and system test routine execution in ten seconds or less. Failures are limited to the plug-in modules and components that affect equipment operation. As here applied, an organizational corrective maintenance action includes the following:

(1) Verification of a fault
(2) Location of the fault to a LRA (Light Replaceable Assembly)
(3) Correction or repair
(4) Check out of the repair

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the currentversion before use.

3.3.7.4.1 Maintenance Concept

(1) By utilization of computer controlled diagnostics and on board test equipment, faults will be isolated to the appropriate WRA or LRA which will then be removed and replaced.

(2) Maintenance replacement of WRAs or LRAs shall be capable of being made in the aircraft using simple hand tools.

3.3.7.4.2 Fault Location Indicators - The fault location indicator for the Receiver-Converter Group shall be a display of the associated display unit under associated computer control.

3.3.7.5 Built-in Test (BIT) Functions - Suitable BIT features shall be incorporated into the Receiver-Converter Group to provide both a failure detection and a fault location function for 95% of the failures when initiated and controlled by the associated computer. (Reference paragraph 3.4.8).

3.3.7.5.1 Built-In-Test (BIT) Operation - The operation of BIT shall be computer initiated with provisions for exercising all functions of the equipment to obtain a GO/NO-GO indication, and for isolating faults to a LRA.

3.3.7.5.2 Built-In-Test (BIT) Indicators - The BIT function shall provide the operator with an indication of equipment status by energizing an advisory or caution indicator (paragraph 3.3.7.4.2) when equipment performance is below an acceptable level, both while on the ground and in flight.

3.3.7.5.3 Receiver-Converter Group Tests - The equipment shall generate and provide test signals to verify the complete operation of the OR-90/ARN-99(V) Receiver-Converter Group when initiated and controlled by the associated computer. The equipment shall provide for the following methods of fault detection and fault isolation:

(1) Systems GO/NO GO Test (SYGNOG)

(2) Diagnostic Tests

3.3.7.5.3.1 Systems GO/NO GO Tests (SYGNOG) - SYGNOG tests performed by the computer shall be primarily a fault detection test to determine functional availability and status of the equipment.

3.3.7.5.3.2 Diagnostic Tests - Diagnostic tests performed by the computer shall provide fault location to the LRAs of the equipment.

3.3.7.6 Special Support Equipment - The Receiver-Converter Group shall not require Special Support Equipment to perform organizational level maintenance actions. General purpose support equipment, i.e., meters, oscilloscopes, will be allowed in the performance of organizational maintenance actions.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the currentversion before use.

3.3.7.7 Packaging Requirements - The Receiver-Converter Group shall be packaged in modular form so that all subassemblies shall be of the plug-in module type, except the Interconnecting Box J-3008/ARN-99(V), and are easily removable without special tools for maintenance.

3.3.7.7.1 Module Arrangement - Module arrangement will be such that access to each module does not require the removal of adjacent modules.

3.3.7.7.2 Keying - Modules and connectors shall be keyed to prevent the insertion of a connector or module into an improper location within or on the equipment.

3.3.7.7.3 Access - It shall be possible to perform 95% of the organizational maintenance actions by access to one side of the Receiver-Converter Group, with the unit still in the aircraft rack.

3.3.7.7.4 Base Plate Pin Configuration - The assignment of functions on the pins of the base plate module plugs, Interconnecting Box J-3008/ ARN-99(V), shall be so configured as to provide adequate separation between power and ground pins. The configuration shall facilitate the use of meter and oscilloscope probes without danger of short circuits. This requirement shall be considered a design goal.

3.3.7.7.5 Loose Hardware - Non-captive hardware (screws, nuts, washers) shall not be used in those areas where organizational maintenance may be required.

3.3.7.7.6 Aircraft Connectors - Aircraft cables shall mount directly to the back panel of the equipment using quick disconnect type connectors.

3.3.7.8 Test Point Requirements - Test points shall be provided to facilitate rapid localization of malfunctions and to allow efficient maintenance, repair, and checkout of the Receiver-Converter Group. Test point definition found in MIL-STD-415 shall apply. The external test points shall be provided on the equipment to measure voltages for use at the organizational level of maintenance when other provisions have not been made for fault isolation to the LRA (module) level. (Test points shall not be used for sensitive signals which could be degraded by the additional wire required to connect the signal to the test point.)

3.3.7.9 Replaceable Assemblies - For the purpose of fault isolation, the WRA's (Weapon Replaceable Assemblies) are:

(1) Antenna-Coupler
(2) Receiver-Converter Group

For the purpose of fault isolation, the LRA (Light Replaceable Assemblies) are defined as the plug-in modules of the Receiver-Converter Group.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

3.3.7.10 Maintainability Verification - Test requirements shall be established under Quality Assurance provisions of this specification as specified in paragraph 4.2.2.7.

3.3.8 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($25^{o}C \pm 10^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 VAC 400 ± 5 Hz and 27.5 ± 0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.10.1 Vibration - The Receiver-Converter Group without isolators shall meet the requirement of Curve I of MIL-E-5400 except that the amplitude shall not exceed 3G (Peak). The Antenna-Coupler shall meet the requirements of Curve IV of MIL-E-5400.

3.3.10.2 Temperature-Altitude - The Receiver-Converter Group shall perform as specified in Specification MIL-E-5400 except that intermittent operation shall be required at temperatures only $4^{o}C$ above continuous operation temperatures.

3.3.11 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 180 seconds under standard conditions, and 300 seconds at extreme service conditions. Equipment shall meet all performance requirements of paragraph 3.4 after this time.

3.3.12 Input Electrical Power

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts:

(1) AC Power (Single Phase), 400 Hz, 115V, Category B, 85 VA, 67 Watts
(2) DC Power, 28V, Category B, 55 Watts warm-up, 5 Watts operating

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

3.3.12.2 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electrical system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.12.3 Warm-up Power - Warm-up power for oscillator heater shall not exceed 2 amps at 27.5 volts dc.

3.3.12.4 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. Fuses or other similar protective devices shall not be used within the equipment.

3.3.12.5 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.13 Cooling - The Receiver-Converter Group and Antenna-Coupler shall operate under all service conditions in accordance with Specification MIL-E-5400 Class 2 (except for modifications in paragraph 3.3.10) as enclosed units without the aid of an active cooling system. Control of the internal temperature rise, and its distribution throughout the Receiver-Converter Group, shall be achieved by conduction of heat from the power dissipating components to the outer finned surface of the case, and from there to air by natural convection. The equipment shall operate satisfactorily under the thermal conditions as determined from evaluation per MIL-T-23103.

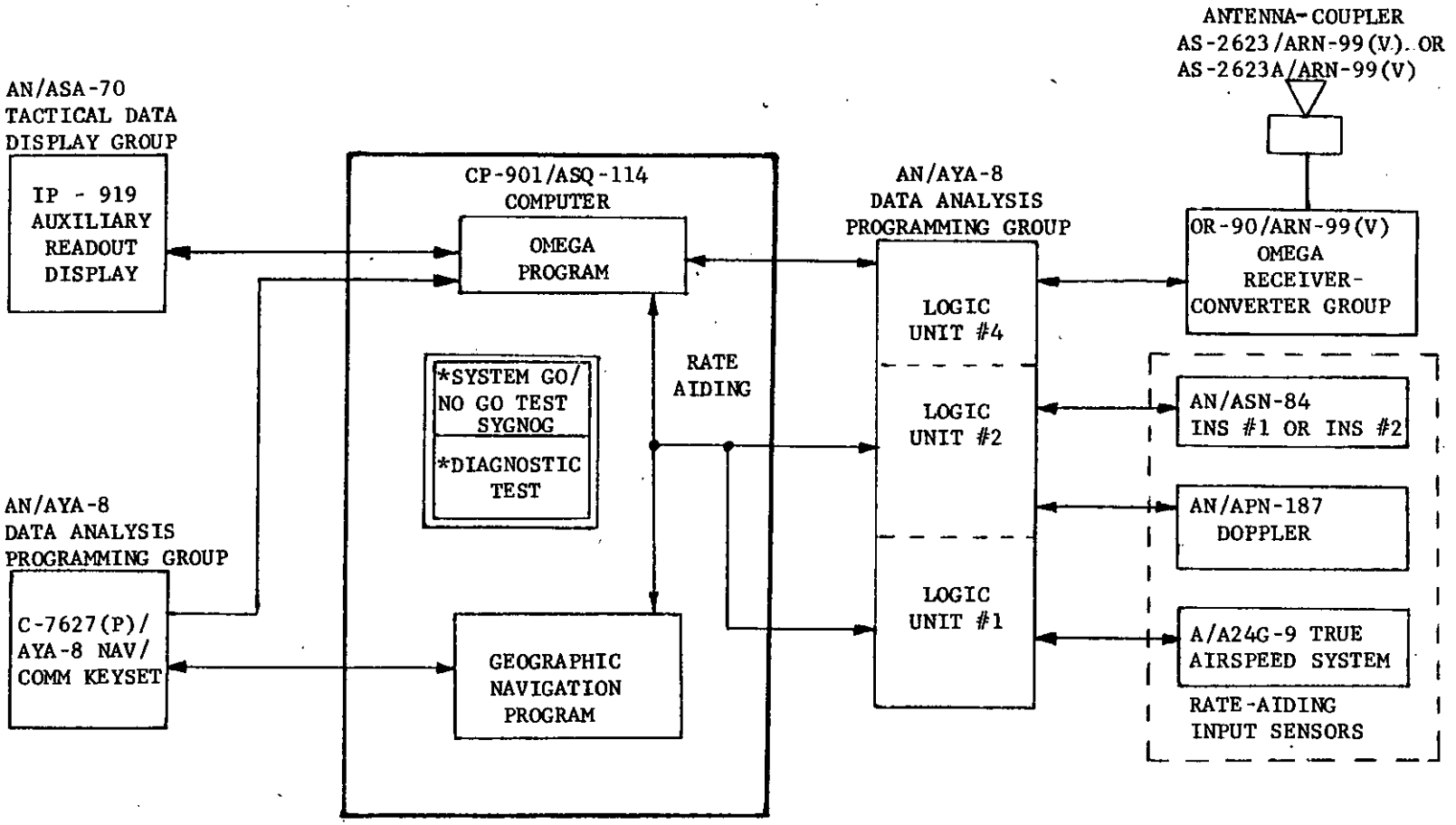
3.3.14 Color - The Receiver-Converter Group shall be lusterless gray, Color Number 36231 in accordance with FED-STD-595. The Antenna-Coupler shall be supplied unpainted.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

3.4.1 Scope of Performance - This section covers the performance requirements of the equipment specified in paragraph 1.2. The performance of the OMEGA Navigational Set, AN/ARN-99(V)1, with its associated equipment, shall be as specified in Appendix A.

3.4.2 Functional Arrangement - The P-3C functional arrangement of the equipment, as part of the OMEGA Navigational Set, shall be as shown in Figure 1. This is a typical arrangement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.



*TEST PROGRAMS ARE ACTIVATED UPON OPERATOR REQUEST

OMEGA NAVIGATIONAL SET AN/ARN-99(V)1
P-3C FUNCTIONAL ARRANGEMENT
(SHOWN AS TYPICAL ARRANGEMENT)

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 20260126T04:36Z
Check the source to verify that this is the current version before use.

3.4.3 Functional Characteristics - The equipment shall have as its primary functional capability the following:

(1) Station/Frequency Usage - The capability of operation with the OMEGA global system using all available minimal path signals from any of the eight (8) transmitting stations on the three (3) OMEGA frequencies of 10.2 KHz, 13.6 KHz and 11 1/3 KHz as shown in Figure 2.

(2) Worldwide Capability - When operating with a completed VLF OMEGA transmitting system, the equipment shall be capable of operation in all latitudes.

(3) Phase Measurement - Determine the phase of incoming signals with the requisite phase measurement accuracy.

3.4.4 Dynamic Input Conditions - The equipment shall have capability of phase tracking the incoming VLF OMEGA signals with the following characteristics:

(1) Signal Strength - The equipment shall have a total dynamic input range of 100 db. Signal strength at input to antenna shall be between 5 μ volts/meter and 500,000 μ volts/meter.

(2) Noise - The equipment shall be capable of phase tracking input signals whose signal amplitude is no worse than 20 db below an ambient noise level of 50 μ V/meter over a bandwidth of 100 Hz.

3.4.5 Antenna Lobe Selection - The equipment under associated Computer control shall use a receiver input switching matrix to select the optimum loop antenna lobe for each OMEGA broadcast station. The switching matrix shall be capable of selecting the following lobing patterns from the loop antenna. Selection for each received frequency shall be independent and shall remain selected until the next selection is made:

(1) Lobe + A (oriented at -135° from forward end of aircraft centerline)
(2) Lobe + B (oriented at -45° from forward end of aircraft centerline)
(3) Lobe - A (oriented at $+45^{\circ}$ from forward end of aircraft centerline)
(4) Lobe - B (oriented at $+135^{\circ}$ from forward end of aircraft centerline)

(Note: The above assumes that the antenna is located on the bottom side of the aircraft and that + angles are measured clockwise [-angles measured counter clockwise] when viewed from above.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

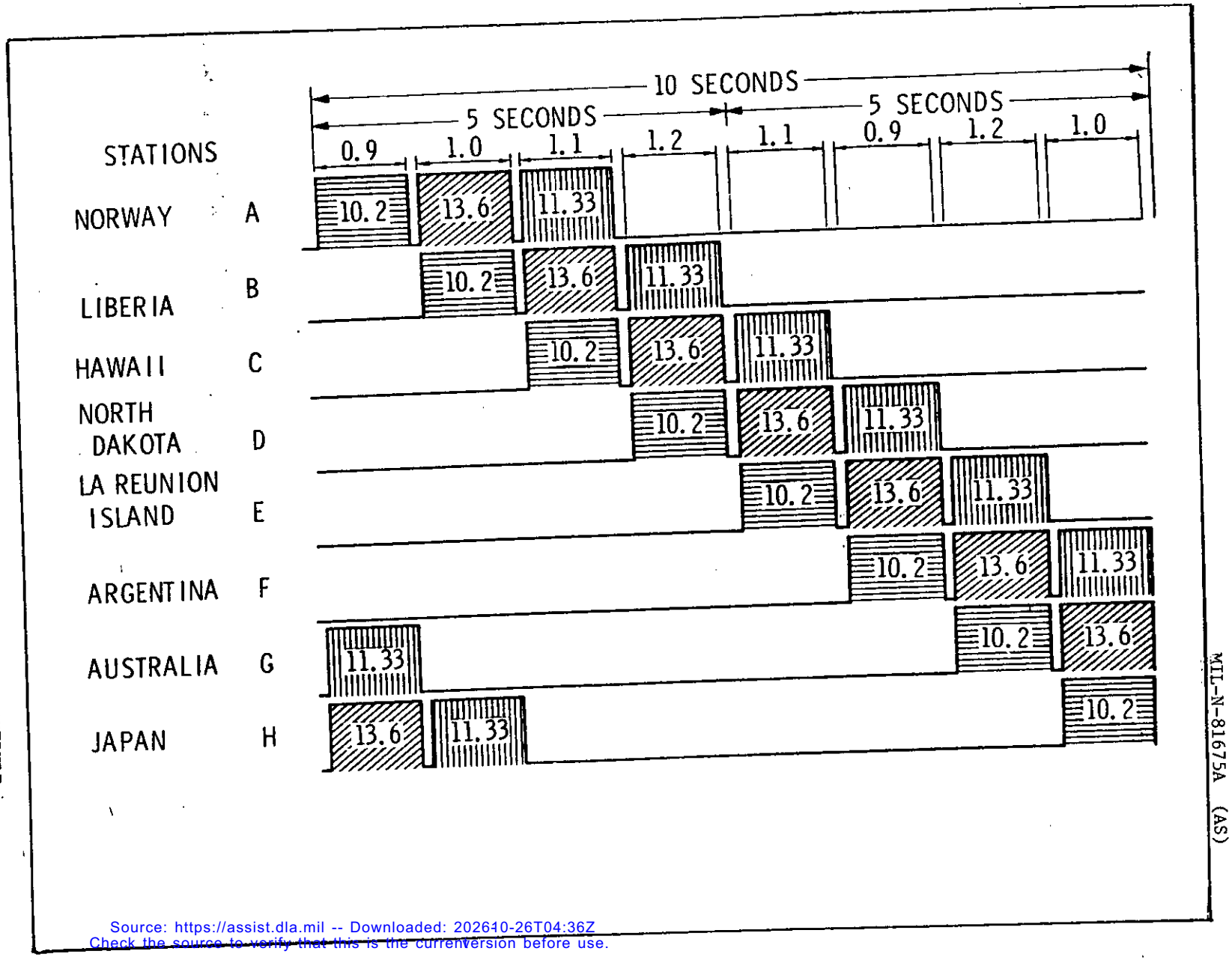


Figure 2 OMEGA TRANSMITTER LOCATION AND BROADCAST SCHEDULE

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the currenversion before use.

3.4.6 Measurement Accuracy - The equipment shall be capable of measuring phase angle (for all OMEGA frequencies and all stations) to within a one standard deviation accuracy of one cec for a dynamic signal input range between 50μ volt/meter and 500,000 μ volt/meter. For a dynamic signal input range betwee 5 μ volts/meter and 50 μ volts/meter, the measurement accuracy shall be within a one standard deviation of two cecs.

3.4.7 Receiver-Converter Group Interface - The Receiver-Converter Group shall interface with the associated computer as described in paragraphs 3.5.1.9 and 3.5.1.10.

3.4.8 Receiver-Converter Group Testing - The Receiver-Converter Group shall perform testing under associated computer control. It is necessary for the computer to command the specific test using the External Function Command Word. The five tests that shall be available are described below. The results of all the tests shall be determined by the computer.

3.4.8.1 Coherence Status - The Coherence Status bit of Data Word No. 1 shall be used only for status of the Precision Frequency Generator's phase coherence. It shall be a logic ONE if the relative phase of the signals out of the Frequency Generator in the Receiver -Converter Group has been disturbed. This test verifies the ability of the countdown logic to be reset to the proper coherence status.

3.4.8.2 RF Section Tests - There shall be a RF section test for each of the Radio Receiver Modules. Two signals differing in phase by approximately 90 degrees shall be derived directly from the Precision Frequency Generator. These shall be coupled directly into the input of each switching matrix like normal operational signals. The associated computer RF test program shall measure the phase of each signal in turn with respect to the Precision Frequency Generator and shall test that the difference is 90 degrees ± 22.5 degrees.

3.4.8.3 Phase Angle to Digital Converter Tests - There shall be a phase angle to digital converter test for each of the three frequencies. A +5V dc signal shall be coupled into each converter in place of the output of the IF stage. The sine and cosine counts shall be tested by the associated computer to determine if they are close to the same nominal value of 1056 counts per second (greater than 512 and less than 1600).

3.4.8.4 Phase Counter Test - There shall be a test to verify that the Phase Counter generates counts at the correct rate. A Phase Counter up command shall cause the test word register in the Phase Counter portion of the receiver to be continuously incremented at a rate of 157.85 counts/50 milliseconds. A Phase Counter down command shall cause the test word register to be continuously decremented at a rate of 157.85 counts/50 milliseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.4.8.5 Receiver-Converter Group Calibration Signals- The OMEGA Receiver-Converter Group shall provide test signals which ensure that the Receiver-Converter Group is functioning, and provide for calibration of each Radio Receiver Module. The Radio Receiver Modules shall be calibrated for:

(1) Scale Factor
(2) Bias
(3) Receiver Phase Shift

These tests shall be conducted automatically and without interference to normal operations under computer control, during the nontransmission time in the OMEGA transmissions. The test signals shall be generated and switched within the Receiver-Converter Group as described in paragraphs 3.5.1.7.1.1, 3.5.1.8 and 3.5.1.10.

3.5 Detail Requirements

3.5.1 Receiver-Converter Group OR-90/ARN-99(V) - The Receiver-Converter Group shall meet the following requirements.

3.5.1.1 Functional Requirements - The Receiver-Converter Group shall perform the basic functions of:

(1) Antenna lobe selection
(2) Generation of precision frequencies for receiver local oscillator and calibration functions
(3) Processing of the three separate OMEGA frequencies
(4) Conversion of receiver outputs to digital format

3.5.1.2 Form Factor - The Receiver-Converter Group shall be enclosed in a conductively cooled package with overall dimensions not to exceed 11.13 inches by 12.66 inches by 7.70 inches exclusive of handles, connector protrusion, knobs, and other mounting hardware.

3.5.1.3 Weight - The weight of the Receiver-Converter Group shall not exceed 33.2 pounds.

3.5.1.4 Contents - The Receiver-Converter Group shall contain the following functional subassemblies:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

(1)	Receiver, Radio 10.2 kHz	R-1880/URN
(2)	Receiver, Radio 11 1/3 kHz	R-1879/URN
(3)	Receiver, Radio 13.6 kHz	R-1878/URN
(4)	Converter-Phase Correlator	CV-3130/URN
(5)	Register, Message, Computer	MX-8940/ARN-99(V)
(6)	Converter, Digital to Digital	CV-2848/ARN-99(V)
(7)	Power Supply	PP-6612/ARN-99(V)
(8)	Interconnecting Box	J-3008/ARN-99(V)

3.5.1.5 Controls - No external controls shall be located on the components of the Receiver-Converter Group.

3.5.1.6 Electrical Connections - Connections to external functions shall be through the Interconnecting Box, J-3008/ARN-99(V) as described in paragraph 3.5.1.12.5.

3.5.1.7 Radio Receiver Modules

Radio Receiver	10.2 kHz
Radio Receiver	11 1/3 kHz
Radio Receiver	13.6 kHz

3.5.1.7.1 Radio Receiver Module Functional Requirements - Each of the three receiver modules shall be a heterodyne type receiver utilizing inputs from an othogonal loop antenna. The RF tuning range shall be three fixed channels of 10.2, 11 1/3, and 13.6 kHz. The overall bandwidth shall be less than 100 Hz. Each receiver module shall provide RF and IF tuning for one of the respective incoming burst signals. Table I summarizes the signal characteristics for each receiver module. The performance of this equipment shall not be degraded by additional OMEGA signal transmissions. The overall spectrum of the OMEGA signals shall be comprised of time-multiplexed transmissions at 10.2, 11 1/3 and 13.6 kHz as transmitted in turn from eight stations. The selectivity of the receiver shall be such that its performance will not be degraded by any CW signal 500 Hz above or below 10.2 kHz and 13.6 kHz, and 200 Hz above or below 11 1/3 kHz having an amplitude as great as one volt/meter when the OMEGA signals are no greater than 500 μ volts/meter.

TABLE I
RADIO RECEIVER MODULES, SIGNAL CHARACTERISTICS

Band Center Frequency	Local Oscillator Frequency	Notch/IF Image Frequency	Intermediate Frequency
10.2 kHz	11 1/3 kHz	12.466 kHz	1.133 kHz
11 1/3 kHz	10.2 kHz	9.066 kHz	1.133 kHz
13.6 kHz	14.73 kHz	15.866 kHz	1.133 kHz

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.7.1.1 Receiver Input Switching Matrix - The Receiver Input Switching Matrix shall provide each of the three receiver channels the capability for the following inputs:

(1) No Input
(2) Antenna Lobe A
(3) Antenna Lobe B
(4) In-phase Test Signal
(5) Quadrature Phase Test Signal
(6) Antenna Lobe A -90°

Inversion of all incoming signals, including test signals, shall be made in the correlator stage of the Converter-Phase Correlator Module. Switching of the above inputs shall be under external computer control. Reference paragraph 3.4.5.

3.5.1.7.1.2 RF Amplifier - Each RF amplifier shall consist of bandpass filtering and limiting. The final stage shall be a heterodyning mixer down to the IF frequency. Image rejection shall be 75 db. (Min.).

3.5.1.7.1.3 Receiver Noise - The Receiver-Converter Group noise reflected to the Receiver-Converter Group input shall be no greater than 50 μ Vrms when measured in a rectangular bandwidth of 100 Hz.

3.5.1.7.1.4 Overload and Recovery Time - Each receiver RF section shall recover signal acquisition in less than 50 milliseconds after a signal causing saturation is removed.

3.5.1.7.1.5 IF Amplifiers - The IF amplifiers shall be tuned amplifiers operating at a frequency of 1.133 kHz having a bandwidth of 50 Hz or less.

3.5.1.7.1.6 Phase Shift - Each receiver channel shall provide stable operation over 60 db range of carrier amplitude variation with less than ± 0.5 cec phase shift, and over 100 db range of carrier amplitude variation with less than ± 1 cec shift.

3.5.1.7.2 Form Factor - Each receiver module shall have the following overall dimensions:

Length	1.200 ± 0.002 inches
Width	10.124 inches ± 0.010 inches
Height	6.00 inches ± 0.04 inches + 1.0 inch for guide pins

3.5.1.7.3 Weight - The weight of each receiver module shall not exceed 2.2 pounds.

3.5.1.7.4 Contents - The contents of each module shall be:

(1) Plug-in Frame
(2) Radio Frequency circuit card
(3) Intermediate Frequency circuit card
(4) Shield

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.7.5 Interconnection - Each receiver module shall interconnect with the Receiver-Converter Group through Interconnecting Box, J-3008/ARN-99(V).

3.5.1.8 Converter-Phase Correlator Module, CV-3130/URN -

3.5.1.8.1 Functional Requirements - This module shall provide the Precision Frequency Generator, Receiver Input Control functions and Phase to Digital Conversion for each Radio Receiver Module. The Converter-Phase Correlator Module shall contain the oscillator to provide simultaneous phase references for the various OMEGA signals. The characteristics of the oscillator shall be as follows:

(1) The frequency drift of the oscillator under continuous operation shall be chosen such that the Receiver-Converter Group will meet the performance requirements specified herein.

(2) The long-term frequency accuracy shall be better than one part in 10^8 per day, five parts in 10^8 per week. The frequency stability shall be within four parts in 10^7 of the nominal frequency within five minutes of application of power.

(3) The power for the heater circuits for the oscillator is derived from a 28 volt dc aircraft supply.

(4) A manual internal frequency or drift control shall be provided as a service adjustment, the use of which will enable depot level compensatory adjustments for slight variations in the oscillator frequency. This control shall permit adjustments of the oscillator over a range of ± 2 parts in 10^6. The overall oscillator accuracy shall permit signal acquisition after a minimum warm-up time.

The Receiver Input Control functions will be those listed in paragraph 3.5.1.7.1.1. The Phase to Digital Converter stage shall form a real-time product of the 1.133 kHz output from the IF amplifiers and Precision Frequency Generator basic timing circuitry. Each of the three RF channels shall have two outputs; one proportional to the sine of the measured phase and the other proportional to the cosine of the same phase. The phase to digital stage of the Converter-Phase Correlator Module will transfer the digital signals of the sine and cosine of the measured phase to the Computer Message Register Module.

3.5.1.8.2 Form Factor - This module shall have the following overall dimensions:

Length	2.100 inches ± 0.004 inches
Width	10.124 inches ± 0.010 inches
Height	6.00 inches ± 0.04 inches plus 1.0 inch for guide pins

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.8.3 Weight - The weight of this module shall not exceed 4.4 pounds.

3.5.1.8.4 Contents - The contents of this module shall be:

(1) Plug-in frame (Two 1.05 inch frames are pinned together to form the Converter-Phase Correlator Module frame).

(2) Phase to Digital # 1 circuit card (10.2 and 11 1/3 kHz phase to digital converters).

(3) Phase to Digital # 2 circuit card (13.6 kHz phase to digital converter and ±13 volt regulator circuit, reference paragraph 3.5.1.11).

(4) Precision Generator circuit card

(5) Digital Data circuit card

(6) Precision Frequency Oscillator

3.5.1.8.5 Interconnection - The Converter-Phase Correlator Module shall interconnect with the Receiver-Converter Group through Interconnecting Box, J-3008/ARN-99(V).

3.5.1.9 Register, Message, Computer Module, MX-8940/ARN-99(V)

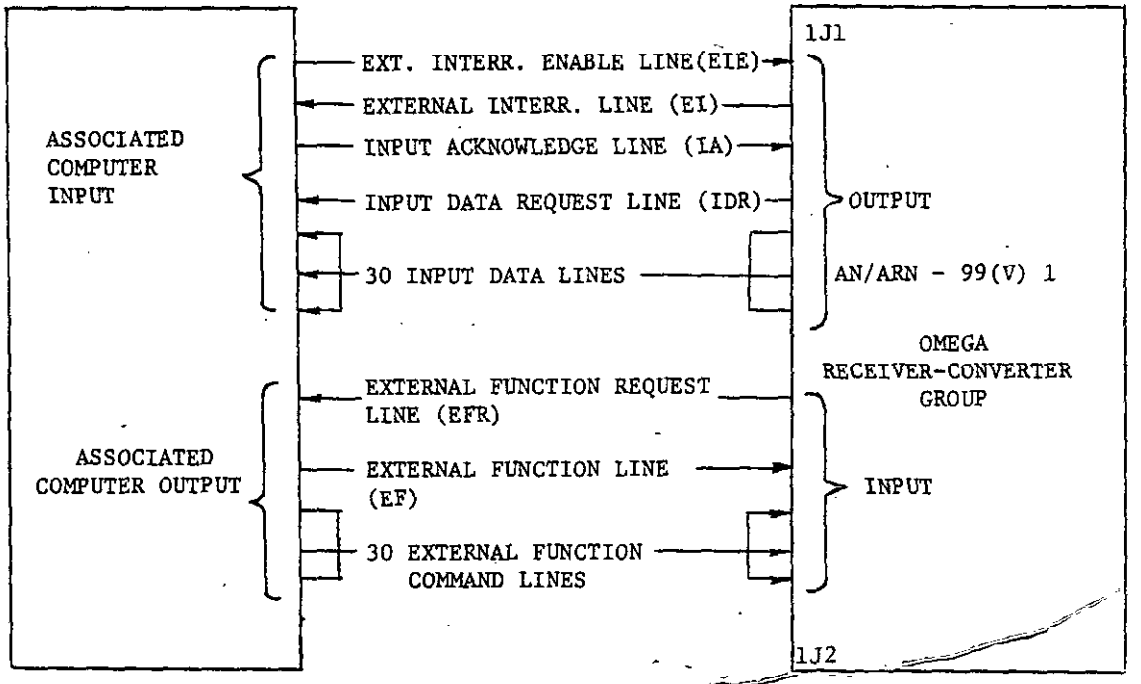
3.5.1.9.1 Functional Requirements - The Computer Message Register Module shall provide for the Receiver-Converter Group output interface control logic and data format, and transmittal of phase data, coherence and data transfer status bits to the associated computer.

3.5.1.9.1.1 Interface Control Logic - The interface control and data lines are shown in Figure 3. The following sequence describes the operation between the Register Module and the associated computer for Receiver-Converter Group output:

(1) The computer, in expectation of the 50 millisecond timing signal from the Register Module, sets the Interrupt Enable (IE) line when OMEGA data is desired.

(2) The Register Module shall detect the IE, place an interrupt word on the computer input data lines, and at the 50 millisecond time interval shall set the Interrupt line and halt the UP-DOWN counters.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.



OMEGA RECEIVER-CONVERTER GROUP
DIGITAL INTERFACE BLOCK DIAGRAM

FIGURE NO. 3

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

(3) The computer detects the Interrupt, and at its convenience, accepts the interrupt word. The computer clears the IE and sets the Input Acknowledge (IA). The Register Module shall detect the setting of the IA, the clearing of the IE and then clear the Interrupt, set the data selector to the first pair of sine and cosine converters, and in less than 5.0 μsec, set the Input Data Request (IDR) line.

(4) If data is not required, the computer does not output the IA and the Register Module clears the IDR after a period of 5 milliseconds.

(5) If data is required, the computer reads the data lines and outputs the IA.

(6) The Register Module shall detect the IA from the computer, clear the IDR, set the data selector to the second pair of sine and cosine counters and reset the IDR line.

(7) The computer reads the second data word and outputs the IA.

(8) The Register Module shall detect the IA, clear the IDR, set the data selector to the third pair of sine and cosine counters, and reset the IDR line.

(9) The computer reads the third data word and outputs the IA.

(10) The Register Module shall detect the IA and clear the IDR.

(11) The Register Module shall continue to initiate the data transfer with Interrupt, only if the Interrupt Enable is set by the computer.

(12) The Register Module shall continue to initiate the data transfer with Interrupt every 50 milliseconds, independent of the associated computer's Input Acknowledge.

(13) The Register Module shall clear any Interrupt or IDR which remains unacknowledged after a period of five milliseconds, +.5, -.2 milliseconds, after being set.

3.5.1.9.1.2 Interface Timing - The Register Module/associated computer interface waveforms shall be per MIL-C-81332.

3.5.1.9.1.3 Word Formats and Description - Every 50 milliseconds the Register Module shall make available to the associated computer one Interrupt Word and three 30 bits Data Words. The words shall be transmitted in parallel format in the following sequence: Interrupt Word, Data Word No. 1, Data Word No. 2, and Data Word No. 3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

The Interrupt and Data Word formats are shown in Figure 4. Transmission of positive values, designated by a logic ZERO in the sign bit space, will be in normal binary form. Negative values, designated by a logic ONE in the sign bit, will be transmitted as a two's complement of the data value. The three Data Words contain the sine and cosine of the measured OMEGA signal phase for each of the three frequencies. The phase coherence status, bit 26 of Data Word No. 1, shall be used only for status of the Precision Frequency Generator's phase coherence. It shall be a logic ONE if the relative phase of the signals out of the Precision Frequency Generator in the Converter-Phase Correlator Module has been disturbed. The data transfer status, bits 00 and 01 of the Interrupt Word, shall indicate to the computer the status of Receiver-Converter Group output data transfer. If the previous Interrupt was not acknowledged by the computer both bits 00 and 01 will be set to a logic ONE. If the Interrupt was acknowledged and the Third IDR was not, only bit 00 will be a logic ONE. In all other cases both bits will be a logic ZERO. A Reset Coherence Command from the computer, see paragraph 3.4.8.1, will set both bits to a logic ZERO.

3.5.1.9.2 Form Factor - The Computer Message Register Module shall have the following overall dimensions:

Length	1.050 ± 0.002 inches
Width	10.124 inches ± 0.010 inches
Height	6.00 inches ± 0.04 inches plus 1.0 inch for guide pins

3.5.1.9.3 Weight - The weight of the Computer Message Register Module shall not exceed 2.0 pounds.

3.5.1.9.4 Contents - The contents of this module shall be:

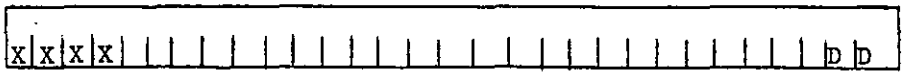
(1) Plug-in frame
(2) Data Selector circuit card
(3) Computer Communication Control circuit card

3.5.1.9.5 Interconnection - The Computer Message Register Module shall interconnect with the Receiver-Converter Group through Interconnecting Box J-3008/ARN-99(V).

3.5.1.10 Converter, Digital to Digital Module, CV-2848/ARN-99(V)

3.5.1.10.1 Functional Requirements - The Digital to Digital Converter Module shall provide for the Receiver-Converter Group input interface control logic, reception of External Function Commands from the associated computer for antenna switching and Receiver-Converter Group testing, and the transfer of commands to the Converter-Phase Correlator Module for the receiver input control functions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.



INTERRUPT WORD

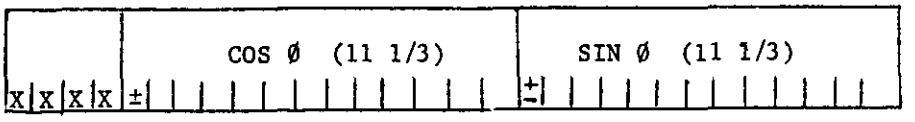


DATA WORD NO. 1

D Data Transfer Status bits
C Coherence Status bit
X Not to be used



DATA WORD NO. 2



DATA WORD NO. 3

OMEGA RECEIVER-CONVERTER GROUP
OUTPUT WORD FORMATS

FIGURE NO. 4

Source: https://assist.dla.mil --.Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.10.1.1 Interface Control Logic - The interface control and data lines are shown in Figure 3. The following sequence describes the operation between the Converter Module and the associated computer for Receiver-Converter Group input:

(1) The Converter Module shall have the External Function Request (EFR) line set indicating that it is ready to accept commands from the computer.

(2) The computer detects the EFR and, at its convenience, places the commands on the 30 External Function Command lines. It then sets the External Function (EF) line indicating the commands are ready for sampling.

(3) The Converter Module shall detect the EF and use it to clear the EFR. The Converter Module then samples the External Function Word.

(4) The computer clears the EF line and then clears the External Function Command lines.

(5) This sequence is repeated when the Converter Module is ready to accept another External Function Command.

3.5.1.10.1.2 Interface Timing - The Converter Module/associated computer interface waveforms shall be as per MIL-C-81332.

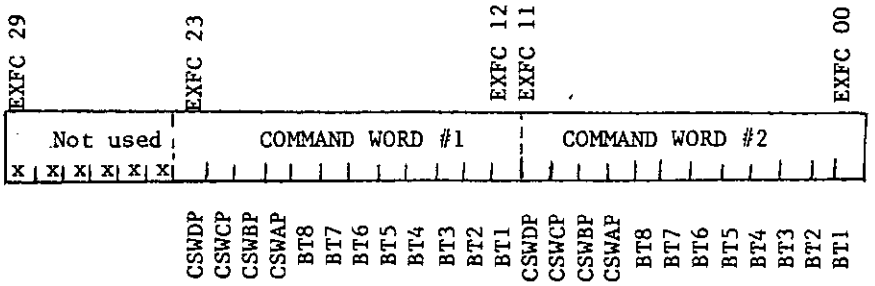
3.5.1.10.1.3 Word Formats and Description - The associated computer must output commands to the Converter Module for antenna switching (reference paragraph 3.4.5) and Receiver-Converter Group testing (reference paragraph 3.4.8). There are two independent commands in each 30-bit parallel transmission. Each command is a 12 bit coded segment of the 30-bit External Function Command. The word format and command coding is specified in Figure 5. Command Word #2 overrides Command #1 when its group select bit CSWAP, CSWBP, CSWCP, or CSWDP is a "1".

3.5.1.10.2 Form Factor - The Digital to Digital Converter Module shall have the following overall dimensions:

Length	1.050 inches ± 0.002 inches
Width	10.124 inches ± 0.010 inches
Height	6.00 inches ± 0.04 inches plus 1.0 inch for guide pins

3.5.1.10.3 Weight - The weight of the Digital to Digital Converter Module shall not exceed 1.7 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.



	CSWDP	CSWCP	CSWBP	CSWAP	BT8	BT7	BT6	BT5	BT4	BT3	BT2	BT1	
10.2 kHz CHANNEL	0	1	0	0	0	0	0	0	0	0	0	1	ANTENNA LOOP A
	0	1	0	0	0	0	0	0	0	0	1	0	ANTENNA LOOP (A-90°)
	0	1	0	0	0	0	0	0	0	1	0	0	ANTENNA LOOP B
	0	1	0	0	0	0	0	1	0	0	0	0	CALIBRATION TEST + 90°
	0	1	0	0	0	0	1	0	0	0	0	0	CALIBRATION TEST
	0	1	0	0	0	1	0	0	0	0	0	0	INVERT ALL FUNCTIONS (180°) (Reference Para. 3.4.5)
11 1/3 kHz CHANNEL	1	0	0	0	0	0	0	0	0	0	0	1	
	1	0	0	0	0	0	0	0	0	0	1	0	
	1	0	0	0	0	0	0	0	0	1	0	0	SAME AS 10.2 kHz CHANNEL
	1	0	0	0	0	0	0	1	0	0	0	0	
	1	0	0	0	0	0	1	0	0	0	0	0	
	1	0	0	0	0	1	0	0	0	0	0	0	
13.6 kHz CHANNEL	0	0	0	1	0	0	0	0	0	0	0	1	
	0	0	0	1	0	0	0	0	0	0	1	0	
	0	0	0	1	0	0	0	0	0	1	0	0	SAME AS 10.2 kHz CHANNEL
	0	0	0	1	0	0	0	1	0	0	0	0	
	0	0	0	1	0	0	1	0	0	0	0	0	
	0	0	0	1	0	1	0	0	0	0	0	0	
RECEIVER DIAGNOSTIC TESTS	V	V	V	V	0	0	0	0	0	0	0	0	NO COMMAND
	V	V	V	V	0	0	0	1	0	0	0	0	RF TEST +90°
	V	V	V	V	0	0	1	0	0	0	0	0	RF TEST
	0	0	1	0	0	0	0	0	0	1	0	0	PHASE ANGLE TO DIGITAL
	0	0	1	0	0	0	0	0	1	0	0	0	PHASE COUNTER UP TEST
	0	0	1	0	0	0	0	1	0	0	0	0	PHASE COUNTER DOWN TEST
	0	0	1	0	0	0	1	0	0	0	0	0	RESET COHERENCE COMMAND (Reference paragraph 3.4.8)

V indicates selection for each frequency or diagnostic (0100, 1000, 0001, or 0010)

OMEGA RECEIVER-CONVERTER GROUP
EXTERNAL FUNCTION COMMAND WORD FORMAT AND CODES
FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.10.4 Contents - The contents of this module shall be:

(1) Plug-in frame
(2) Discrete Storage circuit card

3.5.1.10.5 Interconnection - The Digital to Digital Converter Module shall interconnect with the Receiver-Converter Group through Interconnecting Box J-3008/ARN-99(V).

3.5.1.11 Power Supply Module, PP-6612/ARN-99(V)

3.5.1.11.1 Functional Requirements - The Power Supply Module shall convert the 115V rms 400 Hz single phase input power into regulated dc outputs for the Receiver-Converter Group, OR-90/ARN-99(V), and Antenna-Coupler, AS-2623/ARN-99(V) or AS-2623A/ARN-99(V).

3.5.1.11.1.1 Output Voltages - The Power Supply Module shall provide the following voltages and current capacities with input power of 115V rms 400 Hz, single phase:

+16 ± 0.8 VDC, 0.4 amp
-16 ± 1.0 VDC, 0.4 amp
+5 ± 0.25 VDC, 4.5 amps

The following voltages are regulated by the Converter-Phase Correlator Module, with the listed current capacities. The regulation shall be ± 1% under conditions (1) and (2) in paragraph 3.5.1.11.1.2:

+13 ± 0.7V dc, 0.28 amp
-13 ± 0.7V dc, 0.28 amp

3.5.1.11.1.2 Regulation -The power supply output voltages shall not change more than ±5 percent under the following conditions:

(1) Input line voltage changes ±10 percent
(2) Load changes from no load to full load

3.5.1.11.1.3 Ripple and Noise - The power supply output voltage shall not contain ripple and/or noise voltages in excess of 0.05V rms when measured with a wide band voltmeter.

3.5.1.11.2 Form Factor - The Power supply Module shall have the following overall dimensions:

Length: 2.100 inches ± 0.004 inches
Width: 10.124 inches ± 0.010 inches
Height 6.00 inches ± 0.04 inches plus 1.0 inch for guide pins

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.11.3 Weight - The weight of the Power Supply Module shall not exceed 6.0 pounds.

3.5.1.11.4 Contents - The contents of this module shall be:

(1) The electronics that generate the DC voltages required for the operation of the Receiver-Converter Group.

(2) Plug-in frame (Two 1.05 inch frames are pinned together to form the Power Supply Module frame.)

3.5.1.11.5 Interconnections - The Power Supply Module shall interconnect to the Receiver-Converter Group through Interconnecting Box, J-3008/ARN-99(V).

3.5.1.12 Interconnecting Box, J-3008/ARN-99(V)

3.5.1.12.1 Functional Requirements - The function of the Interconnecting Box shall be to provide the mechanical and electrical interconnections for the plug-in modules of the Receiver-Converter Group, to provide the interconnections from the Receiver-Converter Group to external circuits, and to provide mating provisions for the mounting plate as specified in paragraph 3.5.3. This subassembly shall also contain the time totalizing meter for the Receiver-Converter Group, specified in paragraph 3.3.2.4.

3.5.1.12.2 Form Factor - The overall dimensions of the Interconnection Box shall not exceed, exclusive of front and rear panel handles, fasteners, and connectors:

Length	12.66 inches
Width	11.13 inches
Height	7.70 inches

The width at the plug-in surface shall be 10.125 inches ± .010 inches.

3.5.1.12.3 Weight - The weight of this subassembly shall not exceed 12.5 pounds.

3.5.1.12.4 Contents - The contents of the Interconnecting Box shall be:

(1) Test points
(2) Module connectors
(3) Handles
(4) Nameplate
(5) RFI filter
(6) Time totalizing meter
(7) External connectors

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

(8) Mounting provisions for modules
(9) Front Panel
(10) Back panel
(11) Mating provisions for mounting base

3.5.1.12.5 Interconnections - The Interconnecting Box shall have the following interconnections for external functions:

Reference Designation	Receptacle Type	Function
1J1	M81511/01EF 01A1*	Receiver-Converter Group Output to Computer
1J2	M81511/01EF 01A2*	Receiver-Converter Group Input from Computer
1J3	M81511/02HB02P1	Power Input
1J4	M81511/21EB01S1	Power and Signals (Antenna-Coupler)

*Contacts - MS90461A23-28

3.5.1.12.5.1 I/O Connectors, Pin Designations - The Receiver-Converter Group input data connector 1J2 shall contain the functions and shall be wired as shown in Table III. The Receiver-Converter Group output data connector 1J1 shall contain the functions and shall be wired as shown in Table II.

3.5.2 Antenna-Coupler AS-2623/ARN-99(V) and AS-2623A/ARN-99(V)

3.5.2.1 Functional Requirements - The function of the Antenna-Couplers AS-2623/ARN-99(V) and AS-2623A/ARN-99(V) is to provide for the reception, coupling and amplification of the OMEGA VLF transmissions for the Receiver-Converter Group.

3.5.2.1.1 Characteristics - The characteristics of the Antenna-Coupler are:

(1) Operating frequency - 10 kHz to 14 kHz.

(2) Sensitivity - When subjected to random noise of $7\mu V/M/(Hz)^{1/2}$ the input circuitry and preamplifier noise shall add less than 10% to the resultant noise output.

(3) Sensing elements - Two orthogonal loops

(4) Effective height - Greater than 10 meters on each loop.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

TABLE II

RECEIVER-CONVERTER GROUP OUTPUT DATA CONNECTOR 1J1 CONFIGURATION

FUNCTION	PIN	RETURN
Input Data Request (IDR) (Single Ended)	1	11 (Ground)
Input Data Request (IDR) Complementary)	1	6
Input Acknowledge (1A)	2	7
External Interrupt (EI) (Single Ended)	3	10 (Ground)
External Interrupt (EI) (Complementary)	3	8
Interrupt Enable (EIE)	4	9
Spare*	5	12
D00P	13	21
D01P	14	22
D02P	15	23
D03P	16	24
D04P	17	25
D05P	18	26
D06P	29	39
D07P	30	40
D08P	31	41
D09P	32	42
D10P	33	43
D11P	34	44
D12P	35	45
D13P	36	46
D14P	37	47
D15P	49	58
D16P	50	59
D17P	51	60
D18P	52	61
D19P	53	62
D20P	54	63
D21P	55	64
D22P	56	65
D23P	57	66
D24P	67	75
D25P	68	76
D26P	69	77
D27P*	70	78
D28P*	71	79
D29P*	72	80
Spare*	73	81
Cable Shields	74	

*No functional wiring connected

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

TABLE III

RECEIVER-CONVERTER GROUP INPUT DATA CONNECTOR 1J2 CONFIGURATION

FUNCTION	PIN	RETURN
Output Acknowledge (OA) *	1	6
Output Data Request (ODR) *	2	7
External Function (EF)	3	8
External Function Request (EFR)	4	9
Spare *	5	12
EXFC00	13	21
EXFC01	14	22
EXFC02	15	23
EXFC03	16	24
EXFC04	17	25
EXFC05	18	26
EXFC06	29	39
EXFC07	30	40
EXFC08	31	41
EXFC09	32	42
EXFC10	33	43
EXFC11	34	44
EXFC12	35	45
EXFC13	36	46
EXFC14	37	47
EXFC15	49	58
EXFC16	50	59
EXFC17	51	60
EXFC18	52	61
EXFC19	53	62
EXFC20	54	63
EXFC21	55	64
EXFC22	56	65
EXFC23	57	66
EXFC24 *	67	75
EXFC25 *	68	76
EXFC26 *	69	77
EXFC27 *	70	78
EXFC28 *	71	79
EXFC29 *	72	80
Spare *	73	81
Cable Shields	74	

* No functional wiring connected

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

(5) Antenna pattern - ±45° to half power points on each loop.

(6) E field rejection - Greater than 50 db on each loop.

(7) Null depth - Maximum to minimum ratio greater than 23 db on each loop.

(8) Phase shift - Maximum phase shift change of 5 cec on each loop.

3.5.2.1.2 Power - Power for the Antenna-Coupler shall be +13V dc provided by the Receiver-Converter Group and shall not exceed 0.6 watts.

3.5.2.1.3 Antenna-Coupler Output - The output of each loop shall be two wires, balanced with respect to ground at an impedance of less than or equal to 260 ohms.

3.5.2.1.4 Loop Orientation - The output of the two loops shall be arranged such that the in phase lobe (labeled +A) shall be at an angle of -135° from forward pointing corner, and the other lobe (labeled +B) at -45° from the same forward pointing corner. Referen e paragraph 3.4.5.

3.5.2.2 Form Factor - The form factor of the Antenna-Coupler, exclusive of the connector (including mounting plate thickness) shall not exceed 6.30 inches by 6.30 inches by 2.00 inches centered on a 9.00 ± 0.04 inch diameter circular mounting plate 0.125 inch thick integral with the Antenna-Coupler.

3.5.2.3 Weight - The weight of the Antenna-Coupler shall not exceed 2.85 pounds.

3.5.2.4 Contents - The Antenna-Coupler shall contain a pair of orthogonally mounted ferrite rod loops, each with an integral pre-amplifier. The assembly shall be shielded through use of a Faraday screen. The AS-2623A/ARN-99(V) Antenna-Coupler in addition to the foregoing has a spark gap across each pair of inductors to reduce the possibility of damage to the Antenna-Coupler from high energy inputs caused by lightning.

3.5.2.5 Interconnection - The Antenna-Coupler shall interconnect with the Receiver-Converter Group through the following connector:

Designator	Receptacle Type	Function
2J1	M81511/04HB01P1	Power and Signals

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.3 Mounting Base MT-4534/ARN-99(V) - A mounting base shall be provided to securely accommodate the Receiver-Converter Group. This base is to be affixed at four mounting points to the aircraft equipment rack. The aircraft structure, at the attachment points, shall maintain the relative position of the attachment points under all environmental conditions. The base shall be designed for equipment accessibility (sliding) and removal of the equipment from the base whenever the base is mounted vertically with its left side at the bottom position.

3.5.3.1 Form Factor - The mounting base shall conform to the base dimensions of the Receiver-Converter Group and shall be 15.78 inches by 1.50 inches by 12.08 inches maximum.

3.5.3.2 Weight - The weight of the mounting base shall not exceed 1.8 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facility or any commercial testing laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Test - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of paragraph 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting tests (1), (2) and (3) of paragraph 4.2.2 to the procuring activity for review and approval. The contractor shall submit to the procuring activity a report on tests (4), (5), (6) and (7) of paragraph 4.2.2. This report shall include all pertinent data and observations, major problem areas and recommendations for corrective action as required.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract, and shall include:

(1) Bench Tests
(2) Environmental Tests
(3) Interference Tests
(4) Integration Tests
(5) System Readiness Tests - System GO/NO GO (SYGNOG)
(6) Diagnostic Tests
(7) Maintainability Tests

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

Tests (1), (2) and (3) shall be conducted by the contractor at his facility or any commercial testing laboratory acceptable to the Government. Tests (4), (5), (6) and (7) require the use of government furnished equipment and software programs and shall be conducted at a government designated facility by a designated representative of the government with the assistance of the contractor, if required. Risk of, loss of, or damage to, equipment furnished to the Government for testing shall rest with the Government. Inasmuch as Government testing is not under the control of the contractor, corrective action or changes as required by the Government as a result of Government testing shall be subject to an equitable adjustment in contract price and delivery schedule pursuant to the "Changes Clause" of the contract. However, any damage to, or failure of the equipment which is provided to the Government for their testing that is and can be attributable to the failure of the equipment to meet the design specification, shall be the sole responsibility of the contractor.

4.2.2.1 Bench Tests - Bench tests shall include all tests (at standard conditions) deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract.

4.2.2.2 Environmental Tests - The equipment shall be tested in accordance with MIL-T-5422, except as modified herein. The Sand and Dust test shall be conducted.

4.2.2.3 Interference Tests - The equipment shall be tested in accordance with MIL-STD-461.

4.2.2.4 Integration Tests - The Integration Tests shall consist of a collection of simple subtests designed to establish confidence that the equipment operates according to this specification. The integration subtests shall be constructed to exploit the CP-901/ASQ-114 Computer as a versatile test set. The subtests should be simple in structure requiring minimal dependence on other subtests or their results while at the same time concentrating on as small a functional area as is feasible. The Integration Tests shall provide for the testing of:

(1) Complete input/output interface between CP-901/ASQ-114 Computer and the equipment, including timing, control and data transfer.

(2) Complete functional test of the equipment including all operational functions and all BIT commands and responses.

4.2.2.4.1 Test Control - The control of the Integration Test shall be by an operator via the CP-901/ASQ-114 Computer. The subtest description shall include the input/output words, any required timing tests and/or repetitive loop tests.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

4.2.2.5 System Readiness Tests-System GO/NO GO (SYGNOG) - The SYGNOG Test Program shall be designed to perform a comprehensive GO/NO GO test of all equipment functions. The test shall be designed to complete all test routines in a time not to exceed one minute. Execution of the test in one minute or less is the responsibility of a government designated testing organization.

4.2.2.5.1 NO GO Condition - When an equipment NO GO condition is found, the SYGNOG Test Program shall indicate the fault and shall be designed to lead into the Diagnostic Test Program for further isolation, if the operator so desires. Design of the SYGNOG Test Program is the responsibility of the government designated testing organization.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the currentversion before use.

4.2.2.6 Diagnostic Tests - The Diagnostic Test Program shall be designed to isolate subsystem NO GO conditions to the LRA level. Diagnostic Tests shall be designed to logically follow from NO GO conditions found during SYGNOG testing, but shall not be dependent on SYGNOG. Diagnostic subtests shall be designed to operate without operator intervention whenever possible. Design of the Diagnostic Test Program is the responsibility of a government testing organization.

4.2.2.7 Maintainability Tests - The equipment shall be tested in accordance with MIL-STD-471 as modified by the requirements of paragraph 3.3.7.

4.2.2.7.1 Test Procedures - The following procedure shall be used to demonstrate that the time-to-repair for this equipment is as specified in paragraph 3.3.7.

4.2.2.7.1.1 Test Conditions - One equipment shall be used for this test. Two technicians shall be used to conduct these tests but only one shall be used per repair action as defined in paragraph 4.2.2.7.1.2. Before beginning the test, the equipment shall be checked to determine that it is operating normally. Test equipment, tools, spare parts, Diagnostic Test Programs, and documentation shall be available for the test. The Diagnostic Test Program shall be the primary means of fault detection and isolation.

4.2.2.7.1.2 Repair Actions - A repair action shall be as follows:

(1) Final Fault Isolation after completion of the Diagnostic Test Program.

(2) Repair

(3) Equipment Checkout

4.2.2.7.1.3 Simulated Failures - A total of 30 failures, one at a time, shall be randomly introduced into the equipment. The failures shall be approved by the procuring activity prior to initiation of the test, and shall be limited to those that affect equipment performance.

4.2.2.7.1.4 Inducing Failures - Single failures shall be induced into the equipment by the government representative by substituting modules with known defects. Simulated failures using other means may be used but they shall be kept to a minimum. Each demonstrated failure will consist of a single nonintermittent malfunction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.2.2.7.1.5 Repair Time Measurements - The time required for the technician to perform each repair action, including the use of technical manuals and support equipment, shall be measured starting with final fault isolation after completion of the Diagnostic Test Program and ending at the time the equipment is restored to normal operation and proper operation is verified. Any delay, such as waiting for parts, obtaining test equipment, or administrative delays shall be excluded.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and, for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 2 g's.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be electrically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1*
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

*One out of the first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing Run in Test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrance in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability Production Acceptance Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests," except that all equipments produced shall be tested. Test level E of MIL-STD-781 shall be used.

4.4.3.2.1 All Equipment Tests - Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for 100 hours. Prior to the 100 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the currentversion before use.

(Extend the line as necessary to accommodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the Sampling or Reliability Assurance Tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the Sampling, Reliability Assurance or Special Tests, the following action shall be taken:

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) When practical, include a test in the Individual Test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the Life Tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The Life Test shall be performed under the conditions specified in 4.5.1. The Life Test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for Life Tests). (Test life accumulated during the Reliability Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the table below).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The Life Test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.5.3 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting Preproduction Tests (1 through 7 paragraph 4.2.2), Acceptance Tests and Life Tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary. Procedures for tests 1, 2, and 3 will be generated by the contractor. Procedures for tests 4, 5, 6, and 7 will be in the form of plans in sufficient detail for a Government designated testing organization to convert to detailed procedures and programs.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to Initial Production, Acceptance and Life Tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment defined in this specification is to be used aboard an aircraft to provide present position information to the NAV/COMM operator and Geographic Navigation System.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see paragraph 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.3.8. The correct type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.6 Revisions - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The equipment covered by this specification shall be capable of interfacing with the CP-901/ASQ-114 Computer (containing the OMEGA Program), and using the AN/AYA-8 Data Analysis Programming Group and AN/ASA-70 Tactical Data Display Group for control and display. The performance of the OMEGA Navigational Set, AN/ARN-99(V)1 with the typical associated equipment shall be as specified in Appendix A.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

6.8 Definitions

6.8.1 S/N - The S/N ratio is defined as the ratio of the root mean square voltage amplitude of a continuous sinusoidal signal matching the average amplitude of the signal voltage during an OMEGA transmission to the standard deviation of noise voltage as observed in a rectangular bandwidth of 100 Hz.

6.8.2 "cec" - The "cec" is a phase unit of angular measurement which means "centi-cycle," e.g., 1 cec = 1 percent of a cycle = 3.6 degrees = 4 grades or, approximately, 0.06283 radian.

6.8.3 Bandwidth - The bandwidth is defined for use in this specification as the -3 db bandwidth.

6.9 This specification is under the cognizance of AIR-53353C.

Preparing Activity
NAVY-AS
Project No. 5826-N140

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

APPENDIX A

PERFORMANCE OF NAVIGATIONAL SET, OMEGA AN/ARN-99(V)1 WITH TYPICAL ASSOCIATED EQUIPMENT

10. SCOPE

10.1 Performance - This section specifies the performance of the Navigational Set, OMEGA AN/ARN-99(V)1, with typical associated equipment.

10.2 Components - The Navigational Set, OMEGA AN/ARN-99(V)1 shall consist of the following:

(1) OR-90/ARN-99(V) Receiver-Converter Group
(2) AS-2623/ARN-99(V) or AS-2623A/ARN-99(V) Antenna-Coupler
(3) MT-4534/ARN-99(V) Mounting Base

10.2.1 Associated Equipment - The above components shall be capable of operation with the following equipment:

(1) AN/AYA-8 Data Analysis Programming Group
(2) AN/ASA-70 Tactical Data Display Group
(3) CP-901/ASQ-114 Computer, with OMEGA Navigation Program

20. APPLICABLE DOCUMENTS

20.1 General - The following documents define the program and the operator functions used with the CP-901/ASQ-114 Computer for operation and testing of the Navigational Set, OMEGA, AN/ARN-99(V)1:

(1) OMEGA Navigation, System Function Description NADC Document No. FD-20.

(2) NATOPS Crew Operators Manual (NAV/COMM) - P3C Aircraft.

30. PERFORMANCE

30.1 Functional Characteristics - The OMEGA Navigational Set shall have the following functional capabilities:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

30.1.1 Station/Frequency Usage - The Navigational Set shall have the capability of operation with the OMEGA global system using all available minimal path signals from any of the eight (8) transmitting stations on the three (3) OMEGA frequencies of 10.2 kHz, 13.6 kHz and 11 1/3 kHz.

30.1.2 Automatic Station Synchronization - The Navigational Set shall automatically synchronize to the OMEGA broadcast pattern for the purpose of recognizing and identifying a particular station transmission. The Navigational Set shall remain synchronized to OMEGA broadcast signals during periods of time when the OMEGA signals are not available and the set is operating.

30.1.3 World-Wide Capability - When operating with the completed VLF OMEGA transmitting system, the Navigational Set shall be capable of performing continuous navigation in all latitudes.

30.1.4 Tracking - The Navigational Set shall provide automatic search and lock-on capability for simultaneously tracking up to eight (8) transmitting stations, using three (3) frequencies.

30.1.5 Phase Measurement - The Navigational Set shall determine the phase (relative to the OMEGA broadcast pattern) of incoming signals segregated as to station and frequency, with the requisite phase measurement accuracy.

30.1.6 Rate-Aiding - The Navigational Set shall obtain and use velocity data derived from velocity sensor inputs.

30.1.7 Phase Corrections - The Navigational Set shall automatically provide VLF propagation corrections to the measured phase of the received OMEGA signals for each OMEGA transmitting station on three frequencies.

30.1.8 Position Information - The Navigational Set shall determine the present aircraft position, display this position to the NAV/COMM operator and make this position information available as a selectable update to the Geographic Navigation Program.

30.2 Operation - The OMEGA Navigational Set shall be operated by the NAV/COMM Operator from the NAV/COMM station.

30.2.1 Operational Displays - The display functions of the OMEGA Navigational Set shall be presented to the operator on the IP-919/ASA-70 Auxiliary Readout Display as specified in the P3-C NATOPS Crew Operators Manual (NAV/COMM) (reference paragraph 20.1).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

30.2.2 Operational Controls - Control of the OMEGA Navigational Set shall be through the C-7627(P)/AYA-8, NAV/COMM Keyset as specified in the P3-C NATOPS Crew Operators Manual (NAV/COMM) (reference paragraph 20.1).

30.3 Detailed Functional Characteristics - The OMEGA Navigational Set shall be capable of performing the detailed functions specified in the following paragraphs. The functional program to accomplish these functions is described in the OMEGA Navigation System Function Description (reference paragraph 20.1).

30.3.1 Dynamic Input Conditions - The Navigational Set shall have the capability of phase tracking the incoming VLF OMEGA signals with the following characteristics.

30.3.1.1 Signal Strength - The Navigational Set shall have a total dynamic input range of 100 db. Signal strength at the antenna input shall be between 5μ volts/meter and 500,000μ volts/meter. The navigation accuracy shall not be degraded when any two signals are separated as much as 60 db and no noise is present.

30.3.1.2 Noise - The Navigational Set shall be capable of phase tracking input signals whose signal amplitude is no worse than 20 db below the ambient noise level of 50μ V/meter over a bandwidth of 100Hz.

30.3.2 Antenna Lobe Selection - The Navigational Set under CP-901/ASQ-114 Computer control shall use a receiver input switching matrix to select the optimum loop antenna lobe for each OMEGA broadcast station. The switching matrix shall be capable of selecting the following lobing patterns from the loop antenna. Selection for each received frequency shall be independent and shall remain selected until the next selection is made:

(1) Lobe + A (oriented at -135° from forward end of aircraft centerline)
(2) Lobe + B (oriented at -45° forward end of aircraft centerline)
(3) Lobe - A (oriented at +45° from forward end of aircraft centerline)
(4) Lobe - B (oriented at -135° from forward end of aircraft centerline)

(Note: Reference paragraph 3.4.5)

During the synchronization process, the switching matrix shall electrically rotate the +A signal by -90 so that a psuedo omni configuration of (+ A - 90) + B is used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

30.3.3 Signal Acquisition - The Navigational Set shall be capable of automatic signal acquisition under all signal amplitude conditions, S/N ratios and velocities specified herein. Acquisition time shall not exceed five minutes under any of the signal conditions specified herein. Acquisition is defined as the time required, after the completion of synchronization, to meet the accuracy requirement of paragraph 30.3.8.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

30.3.4 Automatic Synchronization - The Navigational Set shall have the capability of automatic synchronization of the received signals to the OMEGA broadcast pattern to an accuracy of 0.05 second with a resolution of 0.005 second and a confidence factor of 0.9974. Synchronization shall be possible under the following minimum conditions at receiving locations:

(1) Two stations receivable on three frequencies each
(2) One of the two stations shall have a S/N ratio of at least 0 db; the other station shall have a S/N ratio of no less than -10 db on all frequencies.

Maximum time to acquire synchronization under the above conditions shall be less than 180 seconds in 95% of the cases.

30.3.5 Dynamic Phase Tracking - Each tracking filter, when the Navigational Set is operating with an input S/N ratio of -20 db or better, shall meet the following requirements:

(1) The tracking filter lag error shall not exceed one cec in unaccelerated aircraft flight below 200 knots, with no rate-aiding.

(2) The tracking filter shall not lose track in an aircraft subjected to a horizontal acceleration of not greater than 4 ft/sec/sec for not more than one minute.

(3) The tracking filter lag error shall not exceed 3 cecs when an aircraft traveling at 500 knots is subjected to a horizontal acceleration not greater than 45 ft/sec/sec for not more than one minute and the rate-aiding system velocity error is not more than 10%. The Navigational Set shall recover from the effects of the horizontal accelerations in (2) and (3) in less than two minutes after reaching a constant velocity.

30.3.6 VLF Propagation Corrections - The Navigational Set shall automatically provide phase corrections utilizing the present position of the aircraft. These phase corrections shall continuously update the lane position information. The phase corrections shall have at least the following accuracies:

(1) The maximum difference of predicted value minus plot value for any hour of the day shall not exceed 30.0 cecs.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:36Z
Check the source to verify that this is the current version before use.

(2) The mean of the differences over the twenty-four hour periods shall not exceed 18.0 cecs.

(3) The standard deviation of the differences shall not exceed 9.0 cecs.

30.3.7 Measurement Accuracy - The Navigational Set shall be capable of measuring phase angle (for all OMEGA frequencies and all stations) to within a one standard deviation accuracy of one cec for a dynamic signal input range between 50 μ volt/meter and 500,000 μ volt/meter. For a dynamic signal input range between 5 μ volts/meter and 50 μ volt/meter the measurement accuracy shall be within a one standard deviation of two cecs.

30.3.8 Navigation - The OMEGA Navigational Set shall perform the following navigation functions at all latitudes.

30.3.8.1 OMEGA Input Signal Conversion - The Navigational Set shall treat each OMEGA signal burst as a separate measurement and shall provide for conversion of corrected phase signals to a latitude and longitude for the purpose of navigation.

30.3.8.2 Navigation Accuracy - The Navigational Set shall provide geographic position to an accuracy of 1.0 n.mile (rms) daytime and 2.0 n.miles (rms) nighttime when OMEGA input signals are no worse than: Minimum of three frequencies from each of three OMEGA stations, with worst case S/N ratio of -20 db on one station and no worse than -10 db S/N ratios on the other two, and the incoming OMEGA signals are within the predicted propagation limits.

30.3.8.3 Dead Reckoning - The Navigational Set shall provide (DR) dead reckoning navigation from the selected velocity/heading source during the time OMEGA signals are not suitable for OMEGA navigation. No operator intervention shall be required for the Navigational Set to switch from OMEGA navigation to DR navigation and back to OMEGA navigation.

30.3.8.4 Rate-Aiding - The Navigational Set shall use velocity data from the velocity sensors for the purpose of rate-aiding the OMEGA signal phases. Rate-aiding may be automatically derived from any of the following velocity sources:

(1) AN/ASN-84 Inertial Navigation Sets
(2) AN/APN-187 Doppler Velocity Altimeter Radar Set
(3) A/A24G-9 True Airspeed System

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

APPENDIX A

The Navigational Set shall have the capability of deselecting a selected velocity source when a down status output is recognized by the CP-901/ASQ-114 Computer. A secondary velocity source will be automatically selected based upon a priority previously entered into the computer by the NAV/COMM Operator.

30.3.9 OMEGA Navigational Set Self-tests

30.3.9.1 Receiver-Converter Group Testing - The Navigational Set shall perform testing of the Receiver-Converter Group. The CP-901/ASQ-114 Computer shall command specific tests using the External Function Command Word. The tests that shall be available are described below. The results of all the tests shall be dtermined by the computer's examination of the sine and cosine computer inputs and comparison with stored values.

30.3.9.1.1 RF Station Tests - There shall be an RF section test for each of the Radio Receiver Modules. Two signals differing in phase by approximately 90 degrees shall be derived directly from the Precision Frequency Generator. These shall be coupled directly into the input of each switching matrix like normal operational signals. The CP-901/ASQ-114 Computer RF test program shall measure the phase of each signal in turn with respect to the Precision Frequency Generator and shall test that the difference is 90 degrees ± 22.5 degrees.

30.3.9.1.2 Phase Angle to Digital Converter Tests - There shall be a phase angle to digital converter test for each of the three frequencies. A + 5V dc signal shall be coupled into each converter in place of the output of the IF stage. The sine and cosine counts shall be tested by the CP-901/ASQ-114 Computer to determine if they are close to the same nominal value of 1056 counts per second (greater than 512 and less than 1600).

30.3.9.1.3 Phase Counter Test - There shall be a test to verify that the Phase Counter generates counts at the correct rate. A Phase Counter up command shall cause the test word register in the Phase Counter portion of the receiver to be continuously incremented at a rate of 157.85 counts/50 milliseconds. A Phase Counter down command shall cause the test word register to be continuously decremented at a rate of 157.85 counts/50 milliseconds.

30.3.9.1.4 Receiver-Converter Group Calibration Signals - The Navigational Set shall provide test signals which ensure that the Receiver-Converter Group is functioning, and provide for calibration of each Radio Receiver Module. The Radio Receiver Modules shall be calibrated for:

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:36Z
Check the source to verify that this is the current version before use.

(1) Scale Factor
(2) Bias
(3) Receiver Phase Shift

These tests shall be conducted automatically and without interference to normal operations under computer control, during the nontransmission time in the OMEGA transmissions. The test signals shall be generated and switched within the Receiver-Converter Group.

30.3.9.2 Coherence Status - The Coherence Status bit of Data Word No. 1 shall be used only for status of the Precision Frequency Generator's phase coherence. It shall be a logic ONE if the relative phase of the signals out of the Frequency Generator in the Receiver-Converter Group has been disturbed. This status bit can be reset using the Reset Coherence Command.

30.3.9.3 SYGNOG - The SYGNOG Test Program shall be designed to perform a comprehensive GO/NO GO test of all equipment functions. The test shall be designed to complete all test routines in a time not to exceed one minute. When an equipment NO GO condition is found, the SYGNOG Test Program shall indicate the fault and shall be designed to lead into the Diagnostic Test Program for further isolation if the operator so desires.

30.3.9.4 Diagnostic Tests - The Diagnostic Test Program shall be designed to isolate subsystem NO GO conditions to the LRA level. Diagnostic Tests shall be designed to logically follow from NO GO conditions found during SYGNOG testing, but shall not be dependent on SYGNOG. Diagnostic subtests shall be designed to operate without operator intervention whenever possible.

30.4 Definitions

30.4.1 Mean of the Differences - The mean of the differences shall be computed as follows:

$$\bar{d} = \left(\frac{1}{24} \sum_{i=0}^{23} di \right)$$

where di = difference at the ith hour GMT (i=0, 1, 2,---23).

30.4.2 Standard Deviation - The standard deviation shall be computed as follows:

$$= \left[\frac{1}{24} \left(\sum_{i=0}^{23} (di - \bar{d})^2 \right) \right]^{1/2}$$

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/2530

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-N-81675A(AS) Navigation Set, Omega AN/ARN-99(V)1

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:36Z
Check the source to verify that this is the current version before use.